Technology Stack

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Layer	Technology	Version	Role in Platform
Language	Python	3.10 / 3.11 / 3.13	Backend API server, ML pipeline orchestration.
Framework	Flask & Flask-CORS	3.0+	Modular Blueprint routing, REST API endpoints.
Deep Learning	PyTorch & Transformers	2.2+ / 4.38+	Model execution for RT-DETR, CLIP, BLIP-2, Whisper.
Computer Vision	OpenCV (cv2) & Pillow	4.13+ / 10.2+	Video frame decomposition at 1.0 Hz, contour differencing, people / vehicle / object detection, and 3D trajectory projection.
Speech / ASR	OpenAI Whisper	whisper-base	Multilingual speech-to-text with timestamping.
LLM Reasoning	Groq Cloud SDK	LLaMA-3.3-70B	Forensic hypotheses, 4-language translation, Q&A.;
Frontend UI	HTML5, CSS3, ES6 JS	Vanilla ES6+	Cyberpunk forensic HUD, reactive controls, fetch API.
3D Visualization	Three.js & WebGL	r128 / OrbitControls	Interactive 3D crime scene rendering with lighting.
PDF Generation	ReportLab Platypus	4.4.10	Court-admissible PDF synthesis with XML sanitation.